DEEP LEARNING MODULE

Workshop 1

This first workshop aims to give you an opportunity to get used to Python with some simple examples.

1. In the lecture, you saw how we can find the index of the first occurrence of an element.

```
list1 = [13, 45, 33, 67, 18, 56,18,67,67]

print(list1.index(67))
```

would return 3.

Write a function that returns ALL the indices of where the argument occurs. That is, for this example it should return:

[3,7,8]

2. Write and test a function that returns true if a list has duplicates and false otherwise.
3. Try out the following ways of creating numpy (np) arrays:

```
a1 = np.empty((3,2), float)

a2 = np.array([[1,2,3],[4,5,6], [7,8,9]])

a3 = np.ones((3,2))

a4 = np.random.random((2,2))
```

4. Try out the following methods to make sure you understand them:

   Printing the shape of different dimensions of arrays

   slicing arrays

   reshaping arrays

5. Define a class called Module which has the name of the module, the students ids of those taking the module and a dictionary mapping student ids to their marks. Write methods for:

   (i) initialising the class

   (ii) finding the mark achieved by a student

   (iii) working out the average marks obtained by the students.